

# ELECTRONICS (ANALOGUE)

## HWA II

(Understand the construction and principles of operation of solidified power supply)

2005

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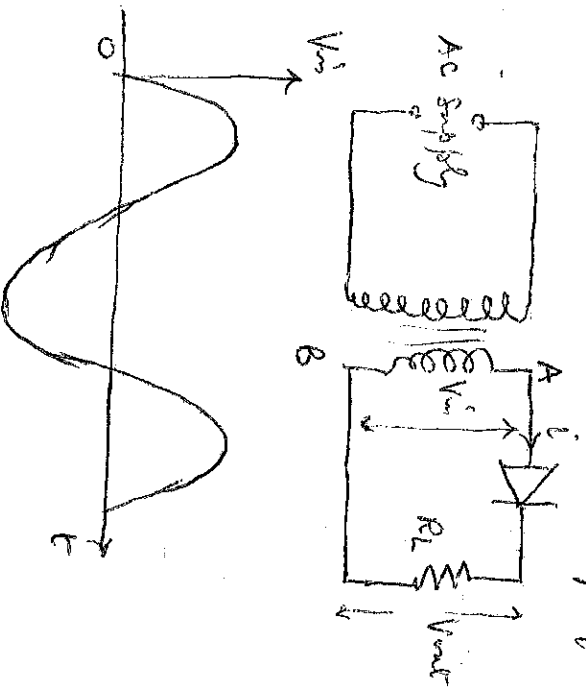
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## Rectification

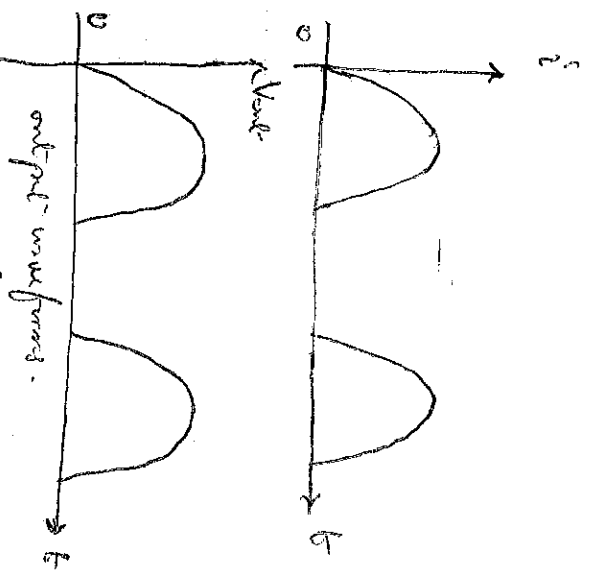
The power by which an alternating current is changed into direct one is called rectification. It can either be half wave or full wave rectification though the use of diodes.

### Half-wave rectification

Below is a circuit which performs rectification (half wave)



input waveform



The AC to be rectified is applied to series with the diode and load resistance  $R_L$ . Usually AC is supplied through transformer so as to step up or down the voltage to the situation demands. The transformer has isolates the rectifier circuit from power line and this reduces the risk of electric shock.

### Principle of operation

The AC voltage across the secondary winding AB changes polarity after every half-cycle. During the positive half cycle, input AC voltage, and A becomes the positive end and B. This makes the diode forward biased and hence it conducts current. During the negative half-cycle, and A is negative end and B. Under this condition, the diode is reverse biased and it conducts no current. Therefore, current flows through the diode during positive half-cycles. In this way current flows through the load  $R_L$  always in the same direction. Hence the output is obtained across  $R_L$ . The output is not steady but pulsating DC. The pulsations can be removed or smoothed out with the help of filter circuits.

## Efficiency of half wave rectifier

$$\eta = \frac{\text{dc output power}}{\text{input ac power}}$$

Let  $V_{ms}$  be the alternating voltage across  $4R$  and let  $r_f$  and  $R_L$  be diode and load resistance respectively. Since output current is pulsating, to find dc power average current has to be found out.

$$\text{Average Current } I_{dc} = \frac{I_m}{\pi}$$

$$\text{where } I_m = \frac{V_m}{R_L + r_f}$$

$$\text{Similarly } V_{dc} = \frac{V_m}{\pi}$$

$$\therefore \text{dc power } P_{dc} = I_{dc}^2 R_L = \left(\frac{I_m}{\pi}\right)^2 R_L$$

$$\text{AC input power } P_{ac} = I_{rms}^2 (R_L + r_f)$$

for half-wave rectified wave

$$I_{rms} = I_m / 2$$

$$P_{ac} = \left(I_m / 2\right)^2 \times (r_f + R_L)$$

$$\therefore \eta = \frac{P_{dc}}{P_{ac}} = \frac{\left(\frac{I_m}{\pi}\right)^2 \times R_L}{\left(\frac{I_m}{2}\right)^2 \times (R_L + r_f)}$$

$$= \frac{0.406 R_L}{r_f + R_L} = \frac{0.406}{\left(\frac{r_f}{R_L}\right) + 1}$$

Efficiency is max when  $r_f$  is negligible as compared to  $R_L$  = 40.6%

actual implies that a max of 40.6% of ac power is converted to dc power.

eg. A crystal diode having  $r_f = 20\Omega$  is used as a rectifier. If the applied voltage  $V = 50\text{sin}\omega t$  and load resistance  $R_L = 800\Omega$ , find (i)  $I_m$ ,  $I_{dc}$ ,  $I_{rms}$  (ii) ac power input and dc power output (iii) dc output voltage and (iv) efficiency of rectification.

Soln:

$$V_m = 50V$$

$$I_m = \frac{V_m}{R_f + R_L} = \frac{50}{20 + 800} = 0.061 A = 61 mA$$

$$I_{rms} = \frac{I_m}{\sqrt{2}} = \frac{61}{\sqrt{2}} = 30.5 mA$$

$$P_{ac} = I_{rms}^2 (R_f + R_L) = (30.5 \times 10^{-3})^2 \times (20 + 800) = 0.763 W$$

$$P_{dc} = I_{dc}^2 \times R_L = (19.4 \times 10^{-3})^2 \times 800 = 0.301 W$$

$$(ii) \text{ dc output voltage} = I_{dc} \times R_L = 19.4 mA \times 800 \Omega = 15.52 V$$

$$\text{or } V_m/\sqrt{2} = 50/\sqrt{2} = 15.52 V$$

$$\eta = \frac{0.301}{0.763} \times 100 = 39.5\%$$

Another way of putting above is called ripple factor defined as  $r = \frac{\text{rms value of alternating component of wave}}{\text{average value of wave}}$

$$\text{It can be shown that for half-wave, } r = \sqrt{\left(\frac{I_{rms}}{I_{dc}}\right)^2 - 1}$$

$$\text{but } \frac{I_{rms}}{I_{dc}} = \frac{I_m/2}{I_m/\pi} = \frac{\pi}{2} = 1.57$$

$$\therefore r = \sqrt{(1.57)^2 - 1} = 1.21$$

Showing that half-wave rectification is a poor way of converting AC into dc.

Peak Inverse Voltage (PIV) <sup>PIV is the maximum reverse voltage that a diode can withstand without destroying the junction and for half-wave  $= V_m$ .</sup>

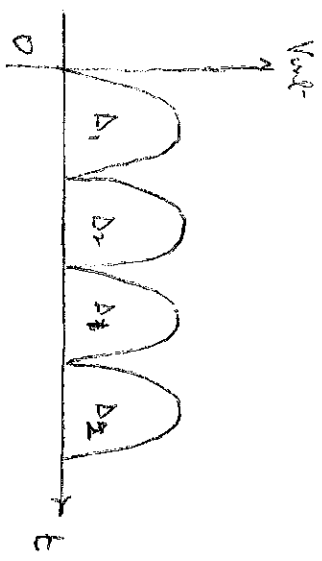
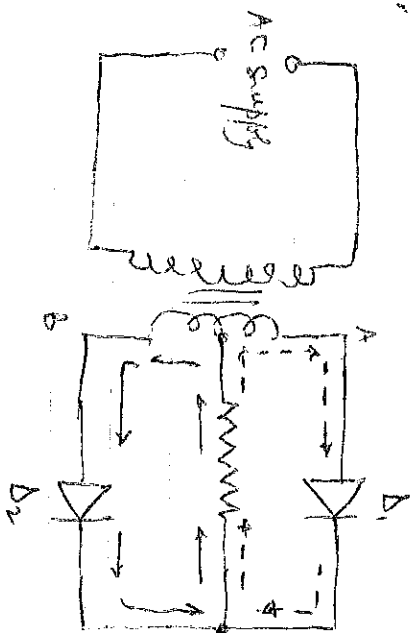
### Full-wave Rectifier

An full wave rectification circuit flows through the load in the same direction for both half-cycles of input ac voltage. The commonly used circuits are:

- Centre-tapped transformer full-wave rectifier.
- Full-wave bridge rectifier.

### Centre-tap full-wave rectifier

It employs two diodes  $D_1$  and  $D_2$  and Centre-tapped transformer as shown below.



operation During the positive half-cycle of secondary voltage, the end A of the secondary winding becomes the end and B -ve.  $D_1$  is forward biased and  $D_2$  reverse biased. So  $D_1$  conducts with conventional current flow through it to the load resistor  $R_L$  shown by dashed lines. During the negative half cycle, reverse above half cycle shown by solid arrow. It can be seen that the load current is in the same direction for both half-cycles of input ac voltage. Therefore the output is obtained across load  $R_L$ .

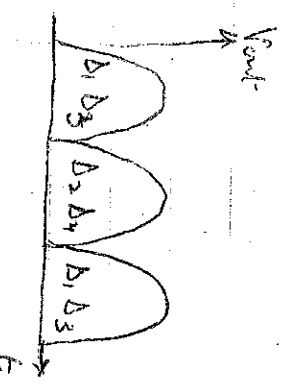
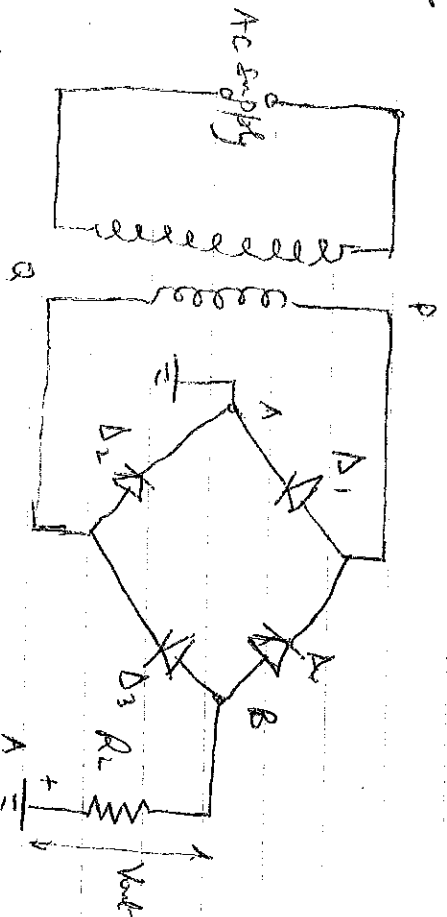
P.I.V. for centre-tapped full wave =  $2V_m$  where  $V_m$  is the max voltage across half secondary winding.

### Disadvantages

- Difficult to locate centre tap on the secondary winding
- Dc output is small as each diode utilizes only one-half of the transformer secondary voltage
- Diodes must have high peak inverse voltage.

### Full-Wave Bridge rectifier

No need for centre tap is eliminated in bridge rectifier. It contains four diodes  $D_1, D_2, D_3$  and  $D_4$  connected to form bridge as shown below



As can be seen above, the need for centre-tap is eliminated

in the bridge. It contains four diodes  $D_1, D_2, D_3$  and  $D_4$  connected to form a bridge.

### Operation.

During the negative half-cycle of the secondary voltage, and  $P$  becomes negative and  $Q$  positive. This makes  $D_2$  and  $D_4$  forward biased and therefore conducts while  $D_1$  and  $D_3$  are reversed biased. During positive half-cycle of secondary voltage, the end  $P$  of the secondary winding becomes positive and end  $Q$  negative. This makes diode  $D_1$  and  $D_3$  forward biased and therefore conducts.  $D_2$  and  $D_4$  are reversed biased. Current now always flows through load  $R_L$ .

PIV PIV of each diode  $\equiv V_m$  of secondary side.

### Advantages

- Centre-tap eliminated
- The output is twice that of the centre-tap for the same secondary voltage.
- For the same output voltage, PIV of bridge circuit is half that of centre-tap circuit.

### Disadvantages

- It requires four diodes
- As during each half-cycle of ac input line diodes that conduct are in series, therefore, voltage drop in the internal resistance of the rectifying unit will be twice as great as in the centre-tap circuit. This is objectionable when the secondary voltage is small.

### Efficiency & full wave rectifier

$$I_{dc} = 2I_m/\pi$$

$$I_{R_{ms}} = \frac{V_m}{\pi + R_L}$$

$$V_{dc} = 2V_m/\pi$$

$$P_{dc} = I_{dc}^2 R_L = (2I_m/\pi)^2 \times R_L$$

(5)

$$I_{rms} = I_m / \sqrt{2} \quad P_{ac} = I_{rms}^2 (r_f + R_L) = (I_m / \sqrt{2})^2 \times (r_f + R_L)$$

$$\therefore \text{Rectifier } \eta = \frac{P_{ac}}{P_{dc}} = \frac{(2 I_m / \sqrt{2})^2 \times R_L}{(I_m / \sqrt{2})^2 \times (r_f + R_L)}$$

$$= \frac{0.812 R_L}{r_f + R_L} = \frac{0.812}{1 + (r_f / R_L)}$$

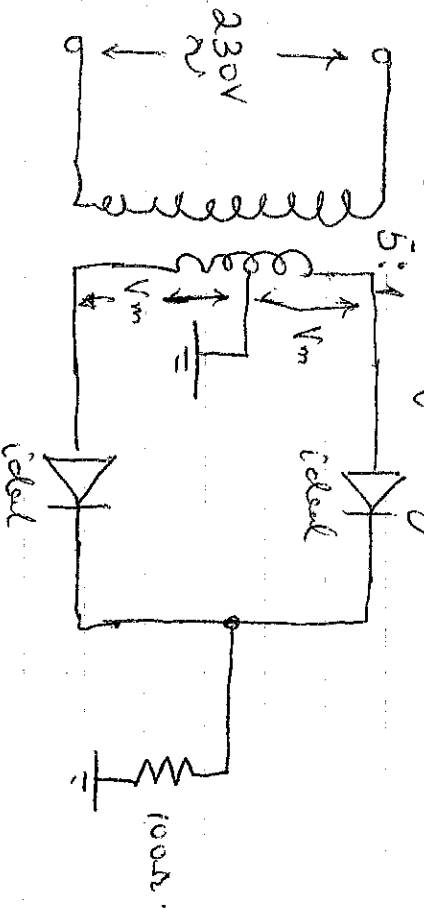
Efficiency 5 max if  $r_f$  is negligible compared to  $R_L$

$\eta = 81.2\%$  which denotes the efficiency due to half-wave rectifier.

### Tutorials

- 1) A full-wave rectifier uses two diodes, the internal resistance of each is assumed constant at  $20 \Omega$ . The transformer rms secondary voltage from Centre-tap to each end of secondary is  $50V$  and load resistance is  $800 \Omega$ . Find
  - (i) rms load current
  - (ii) rms value of load current

- 2) In the Centre-tap circuit shown below, the diodes are assumed to be ideal & having zero internal resistance. Find (i) the output voltage (ii) peak inverse voltage (iii) rectification efficiency.

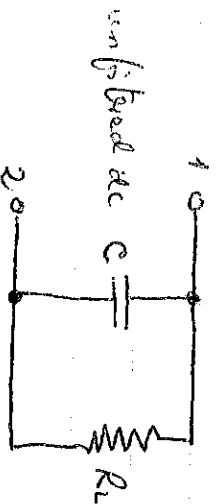


# FILTER CIRCUITS

Generally, a rectifier is required to produce pure dc supply for electronic circuits. However the output of rectifier has pulsating character i.e. it contains ripples superimposed on dc. A filter is therefore required to remove the ripples. The types of filters are Capacitor filter, choke input filter, etc.

## CAPACITOR FILTER

Diagram shown as below:-



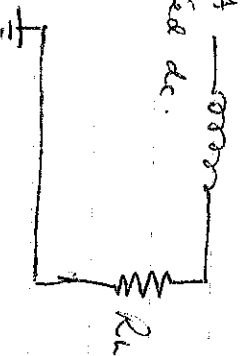
For a full wave capacitor filter,

$$V_{dc} = V_m - \frac{I_{dc}}{4fc}$$

In order to keep the ripple low, very large capacitors must be used. Common ones are electrolytic capacitors.

## CHOKES INPUT FILTERS

Unfiltered dc.



The diagram is as shown above and for a full wave,  $V_{dc} = \frac{2V_m}{\pi} - I_{dc}R$

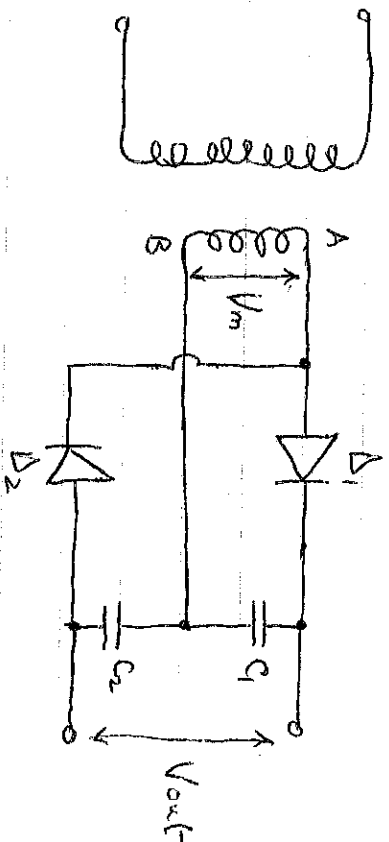
where  $R$  is the total resistance of the circuit (choke, diode and transformer) exclusive of the load.

## MULTIPLYING CIRCUITS

These are circuits which produce high voltage at lower power levels

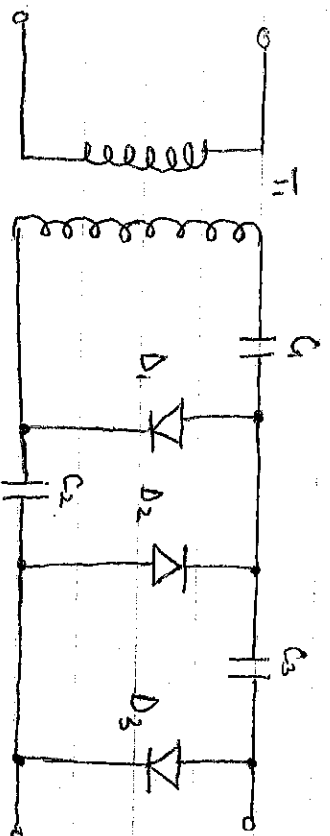
### VOLTAGE DOUBLER

The circuit is given as below:-



The output voltage across  $C_2$  is added to that of  $C_1$   
 $\therefore V_{out} = 2V_m$

### TRIPLER CIRCUIT



The above is a tripler circuit normally used for xrt generation for find anode of tube (TVs, Computer monitors, Radar screen, Oscilloscopes etc).  $T_1$  is a step-up transformer and the output arrangement multiplies the output voltage three times.

## REGULATORS

This is a device which is used to keep the output voltage constant irrespective of changes in the main input voltage and of changes in the load current. These two functions are called line stabilization and load regulation respectively. If the load is to operate properly without drawing excess current, its input voltage need to be kept constant, hence the need for

regulation. The two common methods of achieving stable output voltage is through

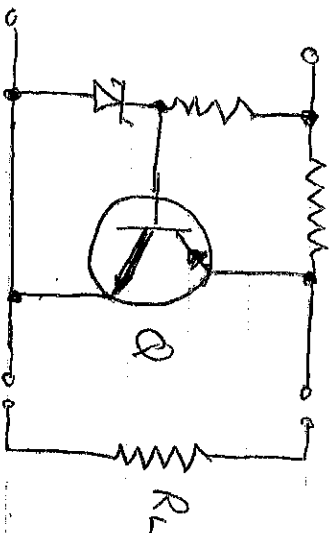
- Linear stabilizer
- Switch mode stabilizer

### LINEAR STABILIZERS

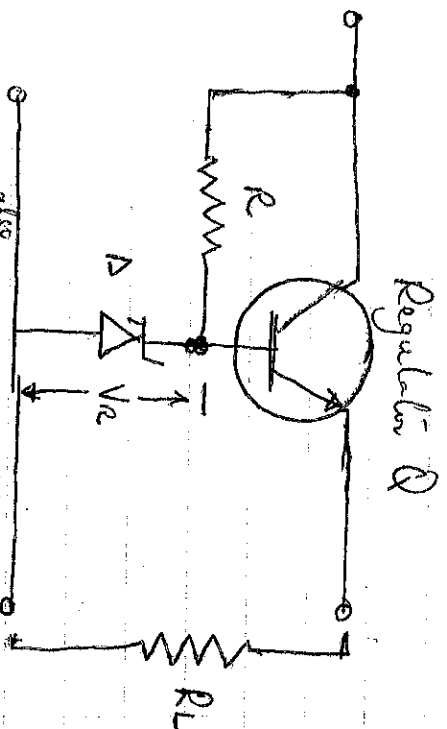
There are of two types:-

- Shunt regulation transistor type
- Series regulation type

Shunt regulation type is given as below

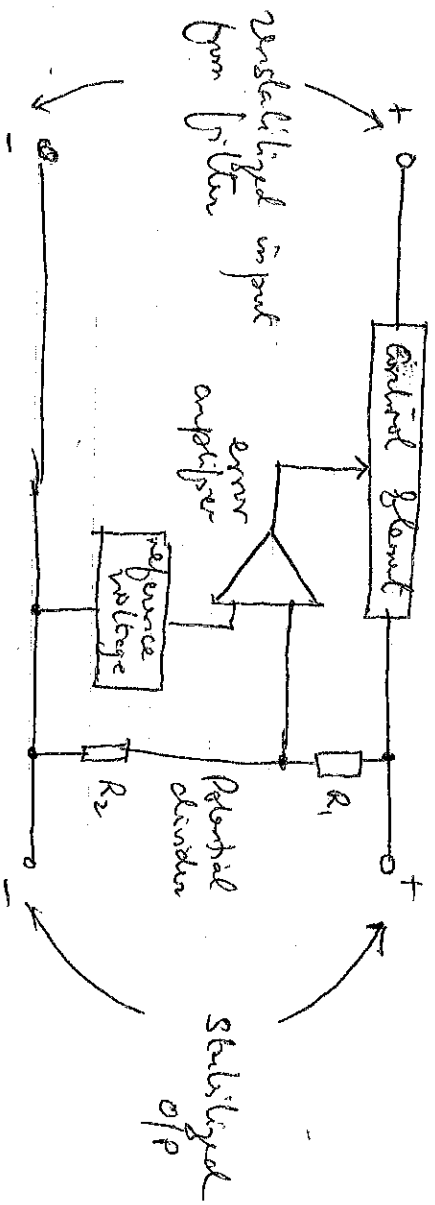


### Series Regulation type



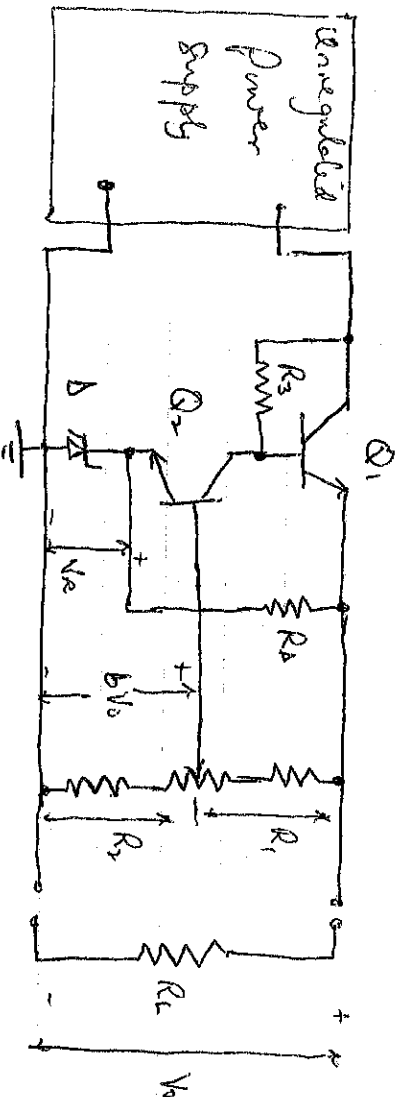
The above is called emitter follower regulation which is an impedance transformation circuit from high to low impedance. A voltage regulation which is given as below which compares the emitter follower is given as below which compares of

- control unit
- reference element (usually a Zener diode)
- an error amplifier



Block diagram of a linear regulator.

### SCHEMATIC DIAGRAM OF LINEAR REGULATOR



Description: The series element or series regulator is  $Q_1$ , the difference amplifier is  $Q_2$  and the reference diode is  $Z$ .

Example: Design a series-regulated power supply to provide a nominal voltage output of 25V and supply load current of  $I_L \leq 1A$ . The unregulated power supply has the following specifications:  
 $V_i = 50$  and  $I_o = 100\text{mA}$

Solution

It can be seen that  $V_o = V_{BEZ} + V_Z + \frac{R_1}{R_1 + R_2} V_o$

$$V_o - \frac{R_1}{R_1 + R_2} V_o = V_{BEZ} + V_Z$$

$$V_o \left[ 1 - \frac{R_1}{R_1 + R_2} \right] = V_{BEZ} + V_Z$$

$$V_o \left[ \frac{R_2}{R_1 + R_2} \right] = V_{BEZ} + V_Z$$

$$V_o = (V_{BE2} + V_E) \left( \frac{R_1 + R_2}{R_2} \right)$$

$$= (V_{BE2} + V_E) \left( 1 + \frac{R_1}{R_2} \right)$$

Here a convenient method for changing the output is adjusting the ratio  $R_1/R_2$  by means of a resistance divider

(rule 3 thumb)

Having established that select a silicon diode with  $V_E \approx V_{E2}$   
Now from 1N755 diodes in series provide  $V_E = 7.5 + 7.5 = 15V$   
and  $R_2 = 120\Omega$  at  $I_2 = 20mA$  (data book)

Choose  $I_{C2} \approx I_{E2} = 10mA$ . The design instructs 2N930

silicon transistor can provide the collector current of  $10mA$ . For this transistor the manufacturer specifies  $I_{C(max)} = 30mA$  and  $V_{CE(max)} = 45V$ .

Now at  $I_{C2} = 10mA$ , the following parameters were measured  
 $h_{FE2} = 220$   $h_{FE2} = 200$   $h_{FE2} = 800\Omega$

Choose  $I_D = 10mA$  so that  $D_1$  and  $D_2$  operate at  $I_2 = 10 + 10 = 20mA$ .

$$R_D = \frac{V_o - V_E}{I_D} = \frac{25 - 15}{10} = 1K\Omega$$

Now the ratio  $R_1/R_2$  is found from

$$V_o = (V_{BE2} + V_E) \left( 1 + \frac{R_1}{R_2} \right)$$

$$25 = (0.6 + 15) \left( 1 + \frac{R_1}{R_2} \right)$$

$$25 = (15.6) \left( 1 + \frac{R_1}{R_2} \right)$$

$$\frac{25}{15.6} = 1 + \frac{R_1}{R_2}$$

$$\therefore \frac{R_1}{R_2} = \frac{25}{15.6} - 1$$

$$\frac{R_1}{R_2} = 0.6$$

$$\therefore R_1 = 0.6 R_2$$

(11)

$$\text{Now } I_{B2} = \frac{I_{C2}}{\beta_{FE2}} = \frac{10 \text{ mA}}{220} = 45 \mu\text{A}$$

It is required that  $I_1 \gg I_{B2}$ , so  $I_1$  is selected to be  $10 \text{ mA}$   
 So  $V_2 = V_{BE2} + V_R = 0.6 + 15.6 \text{ V}$

$$\therefore R_1 = \frac{V_0 - V_2}{I_1} = \frac{25 - 15.6}{10 \times 10^{-3}} = 940 \Omega$$

$$R_2 \approx \frac{V_2}{I_1} = \frac{15.6}{10 \times 10^{-3}} = 1560 \Omega$$

It 2N1772 silicon pnp transistor for  $Q_1$  is selected,

At  $I_C = 1 \text{ A}$ , the following parameters are measured

$$\beta_{FE1} = 125 \quad h_{FE1} = 100 \quad h_{ie1} = 20 \Omega$$

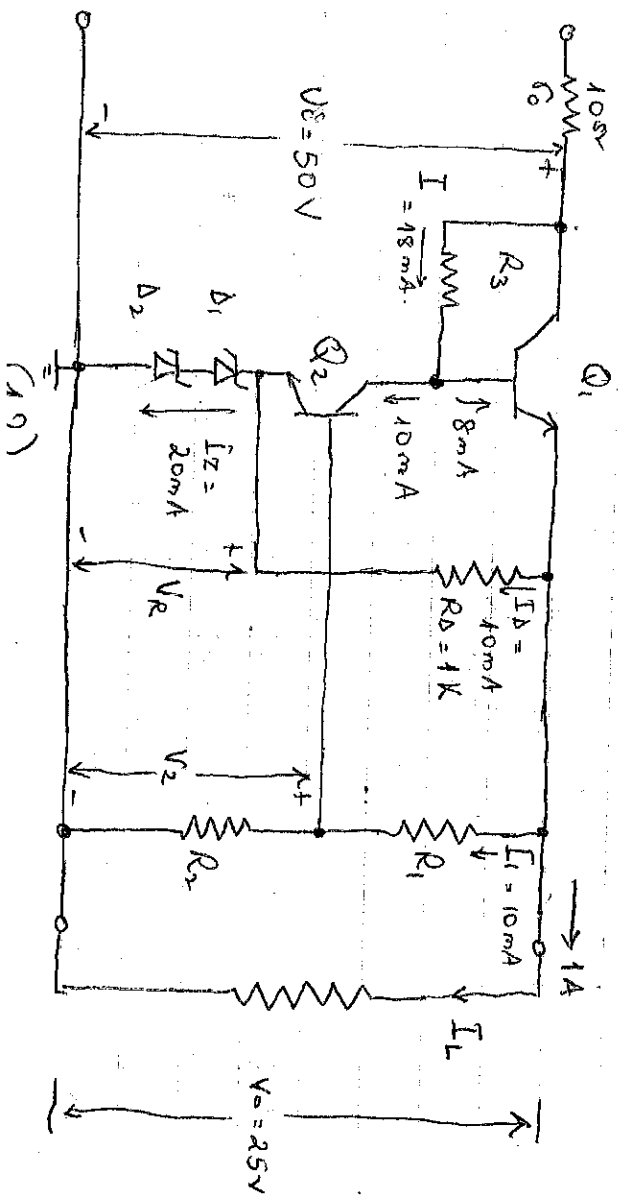
$$I_{B1} = \frac{I_L + I_1 + I_D}{\beta_{FE1}} = \frac{1000 + 10 + 10}{125}$$

$$\approx 8 \text{ mA}$$

The current through resistor  $R_3 = I = I_{B1} + I_{C1} = 8 + 10 = 18 \text{ mA}$ .  
 The value of  $R_3$  to correspond to  $V_i = 45$  and  $I_L = 1 \text{ A}$  is given by

$$R_3 = \frac{V_i - (V_{BE1} + V_0)}{I} = \frac{50 - (0.6 + 25)}{18 \times 10^{-3}} = 1360 \Omega$$

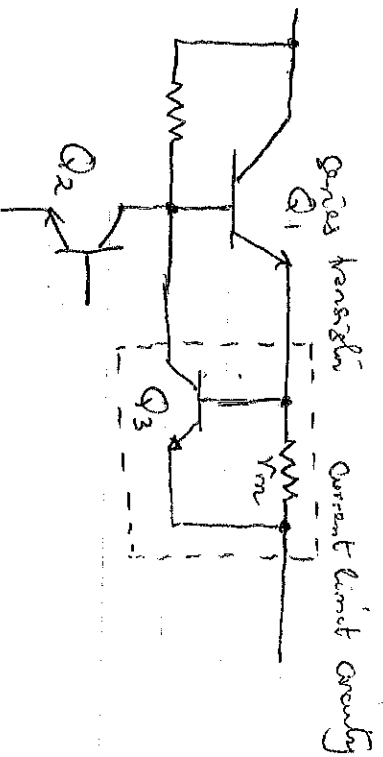
Complete circuit is given as below



## PROTECTION OF POWER SUPPLIES

Protection is - Power Supplies are being provided by fuses on the primary side of transformers. Transformers and diodes on its secondary short circuits before the fuses act unlike semiconductors. A special type of current limiting is employed in case of short circuit and "Crowbar" Circuitry employed in case of overvoltage as shown below:

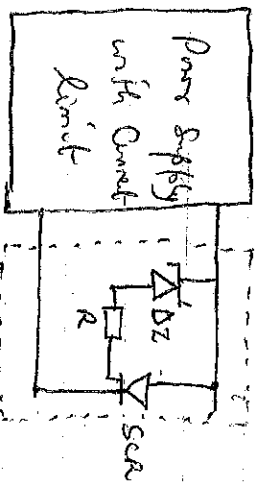
### Current limiter circuit



Power regulation with simple current limiting circuit

operation; the load current flows through a low value current monitoring resistor  $R_m$ . If the load current increases beyond a predetermined value due to short circuit or overload, the voltage develop across this resistor turns on  $T_2$ .  $Q_3$  which is turn leads to turn off  $Q_1$ , the series transistor.

### Overvoltage



This senses the dc output and compares it with reference level. If the dc output voltage rises above  $V_Z$ , a signal is generated which triggers the thyristor and short circuits the output line operating the current limit.

## Design

- a) A full-wave rectifier is to supply a load of 650 mA at 5V dc with a ripple that must be less than 700 mV. Specify the elements of the rectifier using a bridge capacitor and a 78xx voltage regulator integrated chip (Take  $f = 50 \text{ Hz}$  and mains voltage 220V)
- b) Draw the full circuit diagram

### Solution

- a) Choose 7805 IC. 5V, 1A.  
voltage input from 8V to 35 volts  
choose 28 volts as input (dc voltage)

- b) Choosing the working voltage and capacitance of a capacitor  
from  $V_r = \frac{I_{dc}}{4fc}$

$$I_{dc} = 650 \text{ mA}$$

$$V_r = 0.7 \text{ V}$$

Substituting the values yields

$$0.7 = \frac{0.65}{4 \times 50 \times c}$$

$$0.7 \times 4 \times 50 \times c = 0.65$$

$$700c = 0.65$$

$$c = \frac{0.65}{70} = 0.0092857$$

9285.7  $\mu\text{F}$

The standard value and close to the answer is  
and the working voltage is

c) choosing the transformer, from

$$V_m = V_{dc} + \frac{I_{dc}}{4fc}$$

$$V_m = 27 + 0.7 = 27.7$$

$$V_{rms} = \frac{V_m}{\sqrt{2}} = \frac{27.7}{\sqrt{2}} = \frac{27.7}{1.414} = 19.6 \text{ V}$$

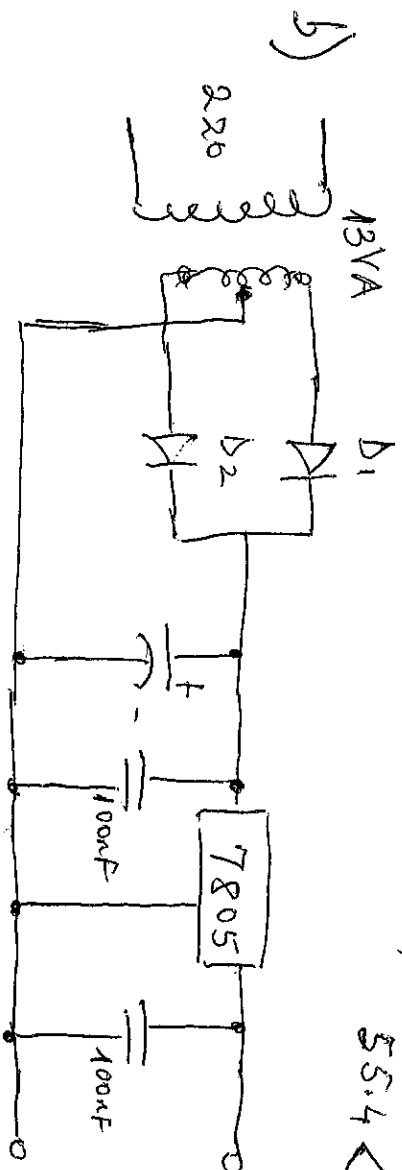
Hence a 18-0-18 would be used (or any slightly higher standard voltage transformer). ~~Choose~~ slightly higher value  
Transformer Specs:  $V = 19 \text{ volts}$  (or any higher standard value)  
 $I = 650 \text{ mA}$

OR

$$V_A = 18 \times 650 \text{ mA} = 11.7 \text{ V}$$

$\approx 13 \text{ V}$  or next slightly

d) 1N4001 chosen. Since  $I = 1 \text{ A} > 650 \text{ mA}$ , PIV =  $2V_m = 2 \times 27.7 = 55.4 < \text{PIV of } 1N4001$   
higher standard value.



# Group A

01 - 10 07 absent  
 Instructions ok.  
 Testing ok  
 Questions ok

V = 240  
 @ 95%  
 V ~ R.  
 12V  
 400Ω

## Group B

11 - 20  
 Instructions ok.  
 Questions ok.  
 Testing ok

$$I = \frac{V}{R} = \frac{12V}{400\Omega} = 0.03A$$

## Group C

22 ok.  
 Instructions ok.  
 Questions ok.  
 Testing ok

24 ok.  
 25 ok.  
 30 ok.

$$R = \frac{V}{I} = \frac{9V}{20mA} = 450\Omega$$

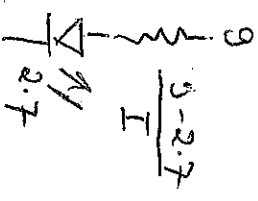
$$0.315 \times 10^3$$

## Group D

31  
 14/10 43  
 Testing ok.  
 Questions ok.  
 Instructions ok

34 ok.  
 39 ✓  
 41 ✓  
 43 ✓

$$9 - 2.7$$



$$\frac{9.27}{20mA} = 315\Omega$$

